Localizing Physical Arbitrariness in the $A_5 \to$ Standard Model Pathway

Masaru Numagaki[1*]

[1]Independent Researcher, Kumamoto, Japan.

Corresponding author(s). E-mail(s): m.numagaki@medcure.co.jp;

Abstract

The gauge group $\mathbf{SU(3)}_C \times \mathbf{SU(2)}_L \times \mathbf{U(1)}_Y$ of the Standard Model with three fermion generations is dissected along the algebraic pathway $\boldsymbol{A_5} \to \mathbf{2}\boldsymbol{I} \to \hat{\boldsymbol{E}}_{\mathbf{8}} \to \boldsymbol{E_6} \times \boldsymbol{A_2} \to \mathbf{SU(3)^3} \to \mathbf{SM}$ under a strict epistemological three-layer separation: machine-verified mathematics (M), physical bridging hypotheses (P), and observational facts (E). All M-layer theorems are formally verified in Lean 4 with sorry = 0, axiom = 0 (approximately 10 800 lines of proof code; see §A for the precise scope of verification). Two main results are established. *Main result 1 (rigidity)*: the core pathway $\boldsymbol{A_5} \to \mathbf{2}\boldsymbol{I} \to \hat{\boldsymbol{E}}_{\mathbf{8}} \to \boldsymbol{E_6} \times \boldsymbol{A_2}$ is uniquely fixed in the M layer with only $\boldsymbol{E_{1a}}$ ($\boldsymbol{n} = \mathbf{3}$) and $\boldsymbol{E_{1b}}$ ($\boldsymbol{N_{\mathrm{gen}}} = \mathbf{3}$) as external inputs. *Main result 2 (localization)*: the minimal hypothesis set $\boldsymbol{\Sigma_{\mathrm{min}}} = \{\boldsymbol{E_{1a}}, \boldsymbol{E_{1b}}, \boldsymbol{E_2}, \boldsymbol{E_3}, \mathbf{P\text{-}4a'}, \mathbf{P\text{-}4b}\}$ is complete and irreducible; the structural P-layer degrees of freedom are localized to exactly two points: the equal-rank democracy principle (P-4a′) and color identification (P-4b).

Keywords: Discrete flavor symmetry, Icosahedral group $\boldsymbol{A_5}$, McKay correspondence, $\boldsymbol{E_8}$ gauge unification, Trinification, Lean 4

1 Introduction

1.1 The problem

The gauge group $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ of the Standard Model (SM), along with three generations of fermions, has been established as an experimental foundation for elementary particle physics. However, the question of *why* this structure is this particular group—the "why this symmetry" problem—remains unresolved.

Existing approaches to this question introduce, to varying degrees, a large number of assumptions. Grand Unified Theories (GUTs) manually select inclusive groups such as SU(5), SO(10), and E_6. Superstring theory leaves a vast number of degrees of freedom for compactification (the landscape problem). Discrete flavor symmetry models arbitrarily introduce finite groups such as A_4, S_4, and $\Delta(27)$. In all cases, the overall picture of the assumptions—what are mathematical consequences, what are physical hypotheses, and what are citations of observational facts—is not always explicit.

The purpose of this paper is to present one answer to the "why this group?" question, using *transparency and minimality of assumptions* as the primary criterion.

1.2 The three-layer epistemological separation

This research program strictly separates all claims in theory construction into the following three layers.

Layer M (machine-verified mathematics).

A group of theorems whose formal proofs using Lean 4 + Mathlib are complete (`sorry = 0`, `axiom = 0`). Purely mathematical facts—such as representation theory of finite groups, combinatorics of Dynkin diagrams, and consequences of the McKay correspondence—belong here. In this paper, the M-layer content is published as approximately 45 files and 10 800 lines of Lean code on Zenodo (DOI: 10.5281/zenodo.19378068). The precise scope of what the Lean type-checker verifies—and what it does not—is discussed in §A.2.

Layer P (physical bridging hypothesis).

A set of hypotheses necessary to connect mathematical structures to physical reality. Each hypothesis is explicitly listed and tagged with **[P]**. The central result of this paper is that the structurally inevitable P-layer hypothesis is localized to exactly two points.

Layer E (citation of observational facts).

Experimental facts cited as theoretical inputs include the existence of a 3-dimensional space ($n = 3$), the number of generations $N_{\text{gen}} = 3$, the non-abelian nature of the strong force, and asymptotic freedom. These are explicitly tagged with **[E]**.

This three-tiered separation is a methodological convention, but its implementation makes the attribution of assumptions at each stage of the theory completely visible.

In addition to the three layers, this paper uses the following two auxiliary attributions.

[M*] (Conditional M-layer theorem).

This designation applies to theorems that close at the M-layer given certain external conditions (a reference to an E-layer or an upstream M-layer theorem). The condition determination process is M, but the motivation for requiring the condition may include **[P]** or **[E]**. For example, M58 (§4.1) includes the condition C3 "family dimension = 3," and C3 is a reference to $N_{\text{gen}} = 3$ **[E]**, so the entirety of M58 is assigned to **[M*]**.

Framework assumption (FA).

The starting point for the entire program, distinct from the individual hypotheses (P layer) within the path. The framework hypothesis FA_1 contains two logically distinct components: FA_{1a} (canonical lift: the McKay output attached to $2I$ uniquely points to affine $\hat{E}_8$) and FA_{1b} (physical adoption: this Lie-theoretic structure is taken as a candidate organizing principle for gauge physics). The first component is mathematically motivated; the second is the genuine framework commitment. Details are given in §2.1. FA_1 stands outside $\Sigma_{\min}$ as a premise, and its non-arbitrariness is discussed in §2.1.

1.3 Overview of the pathway and main results

The pathway discussed in this paper is:

$$A_5 \xrightarrow{\text{double cover}} 2I \xrightarrow{\text{McKay}} \hat{E}_8 \xrightarrow{\text{M58}} E_6 \times A_2 \xrightarrow{\text{M62/M63}} \mathrm{SU}(3)^3 \xrightarrow{\text{P-4b}} \mathrm{SM}. \tag{1}$$

The central problem of this paper is to identify what is mathematically fixed at each stage of this pathway and what still requires physical judgment. To anticipate the conclusion, assuming the framework hypothesis FA_1 (physical development of the McKay correspondence, §2.1), the core path $A_5 \to 2I \to \hat{E}_8 \to E_6 \times A_2$ is uniquely fixed in the M layer with only two empirical inputs: $n = 3$ and $N_{\mathrm{gen}} = 3$. On the other hand, in the final stage $E_6 \to \mathrm{SU}(3)^3 \to \mathrm{SM}$, two points remain as irreducible physical inputs: the equal-rank democracy requirement P-4a′ and the color identification P-4b.

The main findings of this paper can therefore be summarized as follows.

Main result 1 (rigidity).

The route $A_5 \to 2I \to \hat{E}_8 \to E_6 \times A_2$ is uniquely fixed in the M layer with E_{1a} ($n = 3$) and E_{1b} ($N_{\mathrm{gen}} = 3$) as the sole external inputs. This is supported by M49 (McKay–family dictionary), M50–M57 (CP structure), and M58–M61 (E_8 selection and node removal classification).

Main result 2 (localization of physical arbitrariness).

The structural degrees of freedom that remain in the P layer throughout the entire path are localized to two points, P-4a′ and P-4b. In particular, the candidate space for trinification itself is uniquely determined by finite enumeration using M62/M63, but the physical reasons for its adoption and the identification of its color remain outside of algebra.

In this sense, the argument of this paper is not the "derivation of the entire path to the Standard Model" itself, but rather the precise determination of "to what extent mathematics determines the path to the Standard Model, and from what point onward it belongs to physics."

1.4 Original contributions of this paper

In addition to the two main results, the unique contribution of this paper is the trinification stage ($E_6 \to \mathrm{SU}(3)^3$): decomposing the problem into fixing the candidate space

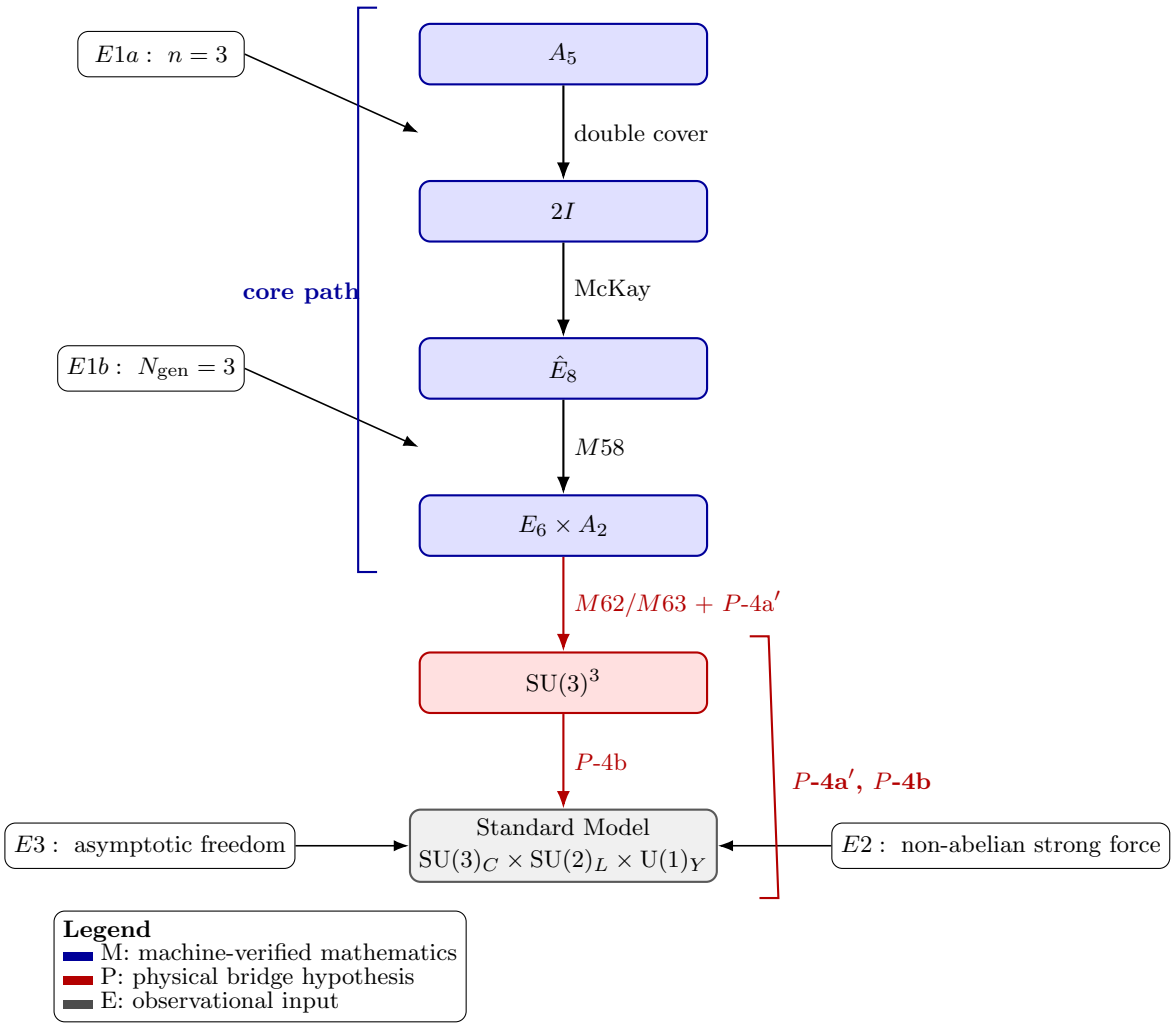

Fig. 1 Vertical overview of the pathway $A_5 \to 2I \to \hat{E}_8 \to E_6 \times A_2 \to \text{SU}(3)^3 \to \text{SM}$. The mathematically fixed core path is shown in blue, while the irreducible physical inputs are localized at P-4a′ and P-4b (red). External observational inputs are $E1a$, $E1b$, $E2$, and $E3$.

(M62/M63, [**M**]) and the physical reason for adoption (P-4a′, [**P**]), and then condensing the content of P. Furthermore, through an audit of all 25 internal hypotheses, we show that Σ_{min} can be reduced to 6 hypotheses (with only 2 structural Ps).

1.5 Structure of the paper

Section 2 discusses the McKay correspondence and family structure (M49). Section 3 discusses the CP structure (M50–M57). Section 4 discusses the uniqueness of $E_8 \to E_6 \times A_2$ (M58–M61). Section 5 discusses trinification selection and P4 three-stage

decomposition (M62–M63). Section 6 discusses the minimal hypothesis set and its epistemological significance. Section 7 discusses comparisons with other theories and future prospects. Section 8 presents the conclusions. Appendix A contains the Lean 4 verification system, Appendix B contains the GAP verification scripts, and Appendix C contains a complete list of P-layer hypotheses.

2 McKay compatibility and family structure

The purpose of this section is to describe the McKay correspondence from the double covering group $2I$ of A_5 to the affine $\hat{E}_8$ Dynkin diagram, and to uniquely identify which irreducible representation of A_5 corresponds to the "3" in $(27, 3)$ within the $E_8 \to E_6 \times A_2$ branch. All content in this section belongs to the M layer.

2.1 The binary icosahedral group $2I$ and the McKay correspondence

A_5 (alternating group, order 60) has five irreducible representations: ρ_1 (trivial, dim 1), ρ_3 (dim 3), ρ_3' (dim 3), ρ_4 (dim 4), and ρ_5 (dim 5), satisfying $\sum d_i^2 = 1+9+9+16+25 = 60 = |A_5|$. The pair ρ_3 and ρ_3' are Galois conjugates, distinguished by their character values on order-5 elements: $\varphi = (1+\sqrt{5})/2$ and $-1/\varphi = (1-\sqrt{5})/2$, respectively.

The Schur multiplier $H^2(A_5, \mathrm{U}(1)) = \mathbb{Z}_2$ gives A_5 a unique double covering group $2I$ (the binary icosahedral group, order 120). $2I$ inherits five representations of A_5 as "A_5-type" and additionally has four "spinorial" representations s_2, s_2', s_4, s_6 (dim 2, 2, 4, 6). With a total of nine irreducible representations, $\sum d_i^2 = 120 = |2I|$ holds.

The McKay correspondence (McKay 1980 [1]; geometric formulation by Gonzalez-Sprinberg–Verdier [2]) asserts that the graph (McKay graph) defined by taking irreducible representations of a finite subgroup $\Gamma \subset \mathrm{SU}(2)$ as vertices and the tensor product decomposition law with the fundamental representation of SU(2) as edges coincides with the corresponding affine Dynkin diagram. For $2I$, the McKay graph coincides with affine $\hat{E}_8$ **[M]**, which is a purely mathematical fact provable by direct computation of representation theory.

Methodological note (*FA_1*).

The output of the McKay correspondence is an affine Dynkin diagram—a graph isomorphic to the classification labels of finite Lie algebras—and not the Lie algebra itself. The correspondence from the Dynkin diagram to a semisimple Lie algebra is unique by Serre's theorem (1966 [3]) **[M]**, and the classification of the E_8 maximal subalgebra (§4), the branching rules of representations, and the Frobenius–Schur indicators used in this paper are all mathematical consequences derived from the structure of this Lie algebra **[M]**.

On the other hand, the step of interpreting the Dynkin diagram output by the McKay correspondence as an algebra of *physical* gauge symmetry involves a judgment that goes beyond the mathematical correspondence. This judgment is explicitly separated as a framework hypothesis with two logically distinct components:

FA_{1a} **(Canonical lift):** Once one seeks a canonical Lie-theoretic structure attached to the A_5 data, the McKay correspondence uniquely selects affine $\hat{E}_8$.

FA_{1b} **(Physical adoption):** This Lie-theoretic structure is taken as a candidate organizing principle for gauge physics.

FA_{1a} is mathematically motivated: the McKay output is unique and canonical, Serre's theorem lifts it to a unique Lie algebra, and no choice is involved. However, the decision to regard this mathematical structure as *physically significant* is not itself a mathematical theorem. FA_{1b} is the genuine framework commitment. The present work studies the physical consequences of adopting this canonical lift; it does not claim that nature must follow this route.

FA_1 differs in nature from the individual bridge hypotheses listed in $\Sigma_{\min}$. While each hypothesis in $\Sigma_{\min}$ is localized to a specific step within the pathway, FA_1 is an explicitly exposed framework choice for unfolding the entire pathway. FA_1 should not be read as a hidden dynamical postulate. Its role is to specify which canonical Lie-theoretic structure attached to the A_5 data is taken as the starting point of the physical analysis. The purpose of this paper is not to prove FA_1 itself, but to determine, once FA_1 is adopted, which parts of the pathway are mathematically rigid and which remain genuinely physical.

Non-arbitrariness of FA_1***.***

The non-arbitrariness of FA_1 is supported by three mutually independent considerations.

(i) Upstream uniqueness of A_5*.* Paper I [4] establishes A_5 as the unique non-solvable finite subgroup of $SO(3)$. If A_5 already occupies a distinguished position upstream, it is natural to examine the canonical Lie-theoretic object attached to it via the McKay correspondence.

(ii) Geometric reappearance. Kronheimer's construction [5] shows that the same ADE package appears independently in the geometry of ALE spaces as hyper-Kähler quotients. This demonstrates that the McKay correspondence is not merely a graph-theoretic coincidence but reflects a deeper geometric structure.

(iii) Historical precedent in string theory. E_8-based algebraic structures recur in orbifold and heterotic string constructions [6]. This does not by itself prove FA_1, but it shows that FA_1 is not an *ad hoc* insertion made solely for the present pathway.

Counterfactual status of FA_1***.***

Table 1 clarifies the consequences of accepting or rejecting FA_1.

If FA_1 is physically meaningful, it should be tested indirectly through the specific pattern of $E_6 \times A_2$ selection (M58), the trinification-based branching relations (§6.7), and the gauge-coupling unification window. In particular, if the generation structure predicted by ρ_3' (M49) is systematically inconsistent with observed flavor data, or if the gauge-coupling running is incompatible with the E_8 Coxeter structure, then FA_1 would be indirectly falsified. Therefore, all conclusions in this paper should be read with the condition "under FA_1".

Table 1 Counterfactual status of FA_1.

FA_1 **adopted**	FA_1 **rejected**
The McKay–E_8 route is read as a physically meaningful pathway.	All M-layer theorems remain valid as pure mathematics; only their physical interpretation is withdrawn.
$E_6 \times A_2$ selection, trinification, and CP structure acquire physical significance.	The pathway $A_5 \to E_8 \to E_6 \times A_2 \to \mathrm{SU}(3)^3$ becomes a mathematical curiosity without physical content.
Experimentally testable consequences arise (§6.7).	No physics claims of this paper are operative.

2.2 Node structure of $\hat{E}_8$ and marks

The affine $\hat{E}_8$ graph is a tree graph with 9 vertices and 8 edges, with the following adjacency structure:

$$0\text{—}8\text{—}7\text{—}6\text{—}5\text{—}4\text{—}3\text{—}1, \qquad 4\text{—}2 \quad \text{(branch)}. \tag{2}$$

The mark (= dimension d_k of the corresponding $2I$ irreducible representation) of each node k is given in Table 2.

Table 2 Marks of the affine $\hat{E}_8$ McKay graph nodes.

Node	0	1	2	3	4	5	6	7	8
Mark	1	2	3	4	6	5	4	3	2

The sum of marks equals the Coxeter number $h = 30$ for E_8 [**M**]:

$$\sum_{k=0}^{8} d_k = 1 + 2 + 3 + 4 + 6 + 5 + 4 + 3 + 2 = 30 = h_{E_8}. \tag{3}$$

Node 4 is the unique trivalent (branch) vertex, with mark = 6 corresponding to the maximum-dimensional spinorial representation s_6 of $2I$. The location of this bifurcation plays an essential role in the parity classification and Galois pair structure described below.
Lean verification: `marks_sum_coxeter`, `branch_point_unique` (E8BlowUp.lean).

2.3 Parity classification

The fundamental node in the McKay graph (node 8, dim 2, the fundamental representation of SU(2)) corresponds to the half-integer spin representation. On the McKay

graph, the nine nodes are separated into two classes—A_5-type (integer spin) and spinorial (half-integer spin)—depending on whether their distance from the fundamental node is even or odd. The affine node 0 is a trivial representation ($j = 0$) and belongs to A_5-type. This parity is completely determined by the topology of the McKay graph [**M**].

Table 3 Parity classification of $\hat{E}_8$ nodes.

Type	Node set	Dimensions	$\sum d_i^2$
A_5-type	$\{0, 2, 3, 5, 7\}$	$\{1, 3, 4, 5, 3\}$	60 ($= \lvert A_5 \rvert$)
Spinorial	$\{1, 4, 6, 8\}$	$\{2, 6, 4, 2\}$	60

The equality $\sum d_i^2 = 60$ for both types reflects that the A_5-type representations correspond exactly to all representations of A_5, and that $|A_5| = |2I|/2 = 60$.

All five A_5-type representations have Frobenius–Schur indicator $\nu_2 = +1$ (real type), and all four spinorial representations have $\nu_2 = -1$ (quaternionic type) [**M**]. This fact provides the starting point for the CP structure in §3.
Lean verification: `A5type_marks_sq_sum`, `spinorial_marks_sq_sum` (A5CP.lean).

2.4 Galois pair {node 2, node 7} = $\{\rho_3, \rho_3'\}$

Among the A_5-type nodes, the only nodes with dim = 3 are $\{2, 7\}$ [**M**]. Since there are only two dim-3 irreducible representations of A_5, ρ_3 and ρ_3', the set correspondence $\{\text{node 2, node 7}\} = \{\rho_3, \rho_3'\}$ is uniquely determined.

Node 2 connects directly to the branch point (node 4) and corresponds to the restriction of the $j = 1$ representation of SU(2) to $2I$. This is a natural 3-dimensional action as an icosahedral rotation of $A_5 \subset \mathrm{SO}(3)$ and identifies with the representation ρ_3 where the character value on the order-5 element is $\varphi = (1 + \sqrt{5})/2$. Node 7 is the remaining dim-3 A_5-type node and identifies with ρ_3' where the character value on the order-5 element is $-1/\varphi = (1-\sqrt{5})/2$. The pair ρ_3 and ρ_3' are exchanged by the Galois automorphism $\sigma\colon \sqrt{5} \mapsto -\sqrt{5}$ (M29 [4]; the complete structure of σ is formulated in §3.4).
Lean verification: `M49_dim3_A5type`, `mckayNode_injective` (A5CP.lean), `M29_galois_permutes_chars` (FlavorGalois.lean).

2.5 M49: McKay–family dictionary

According to M58 (detailed in §4.1), $E_6 \times A_2$ is uniquely selected as the maximal subalgebra of E_8. In the branch from E_8 to $E_6 \times A_2$, the 248-dimensional adjoint representation is decomposed as:

$$248 = (78, 1) \oplus (1, 8) \oplus (27, 3) \oplus (\overline{27}, \bar{3}). \tag{4}$$

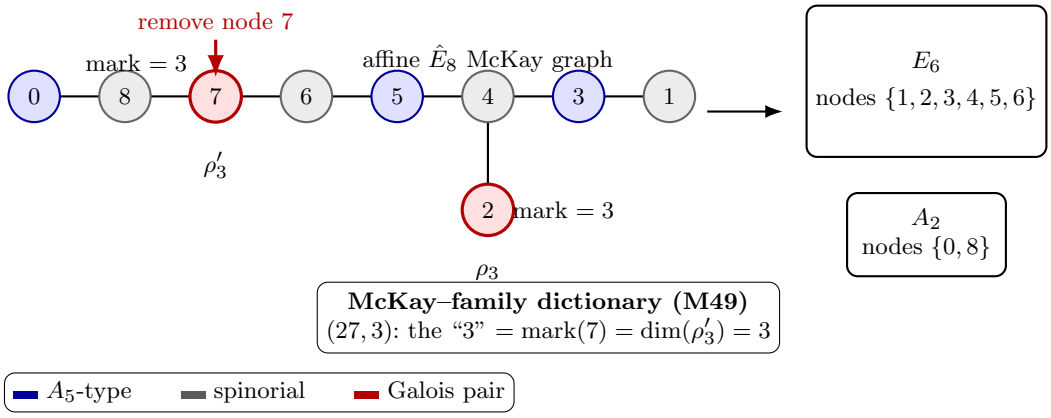

Fig. 2 McKay–family dictionary. The two dim-3 A_5-type nodes form the Galois pair $\{\rho_3, \rho_3'\}$ = {node 2, node 7}. Removing node 7 yields $E_6 + A_2$, and the "3" in $(27, 3)$ is uniquely identified with ρ_3'.

Here the "3" in $(27, 3)$ is $\mathrm{fund}(A_2) = \mathrm{fund}(\mathrm{SU}(3)_{\mathrm{family}})$, the carrier of the family index and the algebraic origin of the 3-generation structure. The purpose of M49 is to identify which irreducible representation of A_5 corresponds to this "3."

In McKay-compatible language, $E_8 \to E_6 \times A_2$ corresponds to the operation of removing node 7 from affine $\hat{E}_8$ (M47, M60 **[M]**). The mark of the removed node 7 is 3, and the residual Dynkin diagram decomposes into $E_6 + A_2$ (M59 **[M]**).

From this structure, the "3" of $(27, 3)$ is determined by the chain: mark of node 7 $\to$ parity classification $\to$ dim-3 A_5-type uniqueness $\to$ ρ_3' identification. Each step is closed at **[M]**, and there is no room for choice. Note that the number of fermion generations = 3 **[E]** ($= E_{1b}$) is referenced as the C3 condition of M58 (§4.1).

Theorem 2.1 (M49: McKay–family dictionary)(a) *By parity classification, the A_5-type nodes are $\{0, 2, 3, 5, 7\}$, and $\sum d_i^2 = 60 = |A_5|$ holds.*

(b) *The only A_5-type nodes with* $\dim = 3$ *are $\{2, 7\}$, forming the Galois pair $\{\rho_3, \rho_3'\}$.*

(c) *Removing node 7 gives $E_6 \times A_2$, and* $\mathrm{mark}(7) = 3 = \dim(\rho_3')$ = *family dimension.*

(d) *The "3" in $(27, 3)$ is uniquely identified as ρ_3' of A_5.*

Lean verification: `mckayNode`, `M49_node7_is_rho3p` (A5CP.lean).

2.6 Galois pair splitting and dictionary rigidity

As a structural consequence of M49, the Galois pair $\{\rho_3, \rho_3'\}$ splits asymmetrically via the $E_8 \to E_6 \times A_2$ bifurcation. ρ_3 (node 2) remains inside E_6 and participates in the gauge structure, while ρ_3' (node 7) is removed and becomes the family index. This asymmetry follows deterministically from the topology of the McKay graph and gives the algebraic origin of the CP structure discussed in §3 (M53–M57).

There is no alternative-by-convention for the M49 identification. Each step of Theorem 2.1 (a)–(d) is uniquely determined in the M layer. The only convention

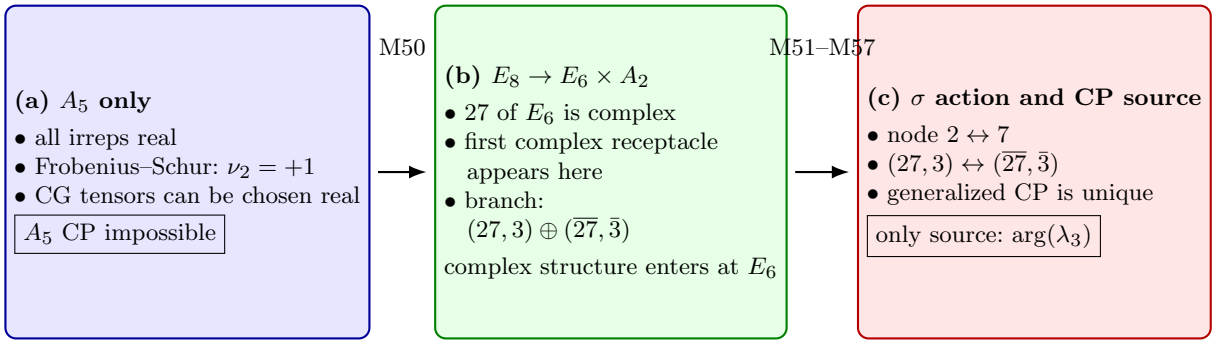

Fig. 3 Three-step summary of the CP structure. CP violation is impossible inside A_5 (M50), a complex receptacle first appears at the $E_8 \to E_6 \times A_2$ bifurcation (M51), and the generalized CP transformation is realized by the Galois involution σ, with unique CP source $\arg(\lambda_3)$ (M52–M57).

dependence is "how to label ρ_3 versus ρ_3'", and the structure as a Galois pair is convention-independent.

3 CP structure: from A_5 to E_6

The purpose of this section is to fully describe the algebraic structure of CP violation in the M layer. To anticipate the conclusion, CP violation is fundamentally impossible inside A_5 (M50), CP enters only as the phase $\arg(\lambda_3)$ of the sole cubic invariant of E_6 (M52), and this structure is uniquely realized as a Galois involution on the McKay graph (M53–M57). Everything in this section belongs to the M layer, provided the $E_6 \times A_5$ framework is assumed.

3.1 M50: A_5 CP impossibility theorem

Theorem 3.1 (M50) *The five irreducible representations of A_5 — ρ_1, ρ_3, ρ_3', ρ_4, ρ_5 — are all of real type (Frobenius–Schur indicator $\nu_2 = +1$).*

The proof is direct. The Frobenius–Schur indicator $\nu_2(\rho) = (1/|G|) \sum_{g \in G} \chi_\rho(g^2)$ can be evaluated in finite computation as a sum over the five conjugacy classes of A_5. For all five representations, we obtain $\nu_2 = +1$.

The condition $\nu_2 = +1$ means, by Frobenius' theorem, that the representation admits a real basis. From this, all representation matrices can be taken real, and by Wigner's theorem (finite group version), the Clebsch–Gordan coefficients can also be chosen real. Therefore, all couplings of A_5-invariant potentials are real, and all rephasing-invariant CP phases vanish:

$$\boxed{A_5 \text{ CP violation is impossible with } A_5 \text{ symmetry alone.}} \tag{5}$$

A complex structure outside of A_5 is essential to generate the CP phase.
Lean verification: `M50_all_real_type` (A5CP.lean, `native_decide`). *GAP verification:* `Indicator(CharacterTable("A5"), [chi], 2)` (M50_M54_verify.gap).

3.2 M51: E_6 fundamental 27 is complex type

Theorem 3.2 (M51) *The fundamental representation* 27 *of* E_6 *is of complex type* ($27 \not\cong \overline{27}$).

The Dynkin diagram of E_6 has a non-trivial $\mathbb{Z}_2$ diagram automorphism, which swaps node 1 $\leftrightarrow$ node 6. This automorphism induces the highest weight exchange $\omega_1 \leftrightarrow \omega_6$. Since $\omega_1 \neq \omega_6$, it follows that $27 = V(\omega_1) \not\cong V(\omega_6) = \overline{27}$.

The comparison between M50 and M51 clarifies the algebraic origin of CP:

Table 4 Comparison of A_5 and E_6 CP structures.

	A_5	E_6
Fundamental rep. type	All real ($\nu_2 = +1$)	27 is complex
Dynkin diagram symmetry	Self-evident (all real)	$\mathbb{Z}_2$ ($\omega_1 \leftrightarrow \omega_6$)
CP phase	Impossible	Allowed

On the path from A_5 to E_6, the complex representation first appears at the bifurcation point $E_8 \to E_6 \times A_2$—that is, the moment when $(27, 3)$ separates from 248. The algebraic origin of the CP phase is precisely localized at this bifurcation point.
Lean verification: `M51_E6_27_isComplex`, `M51_E6_27_ne_27bar` (E6Complex.lean, `decide`).

3.3 M52: Uniqueness of CP source

Theorem 3.3 (M52) *In an* $E_6 \times A_5$*-invariant renormalizable theory (mass term + cubic + quartic), the only independent CP-violating phase is* $\arg(\lambda_3)$ *of the* 27^3 *cubic coupling.*

This theorem consists of four sub-results:

M52a [M]: $\dim \mathrm{Inv}(\mathrm{Sym}^3(27_{E_6})) = 1$. That is, E_6 has exactly one cubic invariant d_{ijk} on 27. The proof begins with the complete decomposition $27 \otimes 27 = V(2\omega_1) \oplus V(\omega_3) \oplus \overline{27}$, where $V(2\omega_1) = \mathbf{351}$, $V(\omega_3) = \mathbf{351}'$, and the dimension is consistent as $351 + 351 + 27 = 729 = 27^2$. Since $\overline{27}$ belongs to $\mathrm{Sym}^2(27)$, the invariant belongs to $\mathrm{Sym}^3(27)$, and its dimension is 1.

M52b [M]: From M50, the A_5-invariant CG tensor can be taken with a real basis.

M52c [M]: The quartic coupling is real by hermiticity. Since $(27 \times \overline{27})^2$ is self-adjoint, the quartic coupling constant can be taken real.

M52d [M]: Under the field rephasing $\Phi \to e^{i\alpha}\Phi$, $\lambda_3 \to e^{3i\alpha}\lambda_3$, but $\arg(\lambda_3)$ cannot be removed because it affects other couplings (mass term, quartic term).

Combining these four sub-results, the sole source of CP violation in the $E_6 \times A_5$-invariant theory is localized to $\arg(\lambda_3)$:

$$\delta_{\mathrm{CP}} = 0 \iff \arg(\lambda_3) = 0 \bmod \pi. \tag{6}$$

The attribution of M52 is $[\mathbf{M}^*]$. If we accept $E_6 \times A_5$ as a framework, the uniqueness of the CP source is a purely mathematical consequence, but the framework $E_6 \times A_5$ itself depends on M58 in §4.

Scope of applicability. M52 holds within renormalizable theory (mass term + cubic + quartic). The possibility of additional CP sources from effective operators of dim ≥ 5 depends on the UV completion and is beyond the scope of this paper.
Lean verification: `M52_unique_CP_source` (A5CP.lean). *GAP verification:* M52a_verify_v2.gap.

3.4 M54: Structure of Galois involution on the McKay graph

Theorem 3.4 (M54) *The Galois automorphism $\sigma\colon \sqrt{5} \mapsto -\sqrt{5}$ acts as an involution on the 9 irreducible representations of $2I$. σ consists of two transpositions:*

$$\rho_3 \leftrightarrow \rho_3' \quad (A_5\text{-type}), \qquad s_2 \leftrightarrow s_2' \quad (\text{spinorial}), \tag{7}$$

with five fixed points $\{\rho_1, \rho_4, \rho_5, s_4, s_6\}$.

σ preserves dimension ($d_{\sigma(i)} = d_i$), parity (A_5-type $\leftrightarrow$ A_5-type, spinorial $\leftrightarrow$ spinorial), and the Frobenius–Schur indicator ($\nu_{2,\sigma(i)} = \nu_{2,i}$).

The A_5-type transposition $\rho_3 \leftrightarrow \rho_3'$ corresponds to the exchange of node 2 $\leftrightarrow$ node 7 on the McKay graph. The fixed nodes ρ_4 (node 3) and ρ_5 (node 5) are σ-invariant because they have rational character values.

The proof is carried out by implementing σ as GaloisCyc$(\cdot, 2)$ in GAP—an automorphism that acts as $\zeta_5 \mapsto \zeta_5^2$ over the cyclotomic field and induces $\sqrt{5} \mapsto -\sqrt{5}$—and directly computing the action on the character values of all 9 representations.
Lean verification: `M54_involution`, `M54_preserves_dim`, `M54_preserves_parity`, `M54_fixed_count` (A5CP.lean, `native_decide`).

3.5 M53, M55–M57: McKay–CP correspondence and channel splitting

From M49 and M54, the following consequence theorems are derived.

Theorem 3.5 (M53: McKay–CP correspondence) *σ: node 2 $\leftrightarrow$ node 7 corresponds to the exchange $(27, 3) \leftrightarrow (\overline{27}, \bar{3})$ and gives the algebraic framework of the generalized CP transformation.*

This correspondence is a direct composition of M49 (node $7 = \rho_3' \to$ family) and M54 (σ swaps nodes 2 and 7). Combined with the fact that removal of node 7 yields $E_6 \times A_2$ (M60, §4.3), σ naturally emerges as a mapping that swaps $(27, 3)$ and $(\overline{27}, \bar{3})$ on the $E_8 \to E_6 \times A_2$ branch.
Lean verification: `M53_sigma_swaps_nodes_2_7` (A5CP.lean).

Theorem 3.6 (M55: σ action on the 248 branching) *σ acts on the 248 branches of $E_8 \to E_6 \times \mathrm{SU}(3)$ as shown in Table 5.*

Table 5 σ action on 248 branch components.

Component	σ action	Reason
$(78, 1)$ adjoint	fixed	adjoint is real type
$(1, 8)$ adjoint	fixed	SU(3) adjoint is real type
$(27, 3)$	$\to (\overline{27}, \bar{3})$	M54: $\rho_3' \leftrightarrow \rho_3$ + M51
$(\overline{27}, \bar{3})$	$\to (27, 3)$	converse

In $248 = 78 + 8 + 81 + 81$, σ fixes the adjoint components (86 dimensions) and exchanges the matter components (162 dimensions) as a complex conjugate pair.
Lean verification: `M55_involution_248`, `M55_swaps_27` (A5CP.lean).

Theorem 3.7 (M56: σ–CP uniqueness) $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q}) = \{\mathrm{id}, \sigma\}$ *is a group of order 2, and its only nontrivial Galois automorphism is σ. Thus, the algebraic structure of the generalized CP transformation is uniquely determined.*

The CP conservation condition is equivalent to all couplings in the potential being σ-invariant (belonging to $\mathbb{Q}$). Combined with M52, CP violation is equivalent to $\arg(\lambda_3) \neq 0 \bmod \pi$.
Lean verification: `M56_unique_sigma` (A5CP.lean).

Theorem 3.8 (M57: CP-sensitive channel splitting) *σ swaps ρ_3 and ρ_3' while fixing ρ_4 and ρ_5. Due to this asymmetry, the three flavor channels of A_5 behave essentially differently in their sensitivity to CP.*

The comparison is summarized in Table 6.

Table 6 CP sensitivity channel splitting.

	Channel II (mass)	Channel III (mixing)
Tensor product	$\rho_3 \otimes \rho_3' \otimes \rho_5$	$\rho_3 \otimes \rho_3' \otimes \rho_4$
CG eigenvalue field	$\mathbb{Q}(\sqrt{5})$	$\mathbb{Q}$
CP character	CP-sensitive	CP-even

In both channels, σ exchanges $\rho_3 \leftrightarrow \rho_3'$, but the key to the splitting lies in the third factor. Since all character values of ρ_4 are rational, the CG eigenvalues of Channel III remain in $\mathbb{Q}$ and are σ-invariant (CP-even), while in Channel II, which involves ρ_5, the Galois asymmetry propagates to the CG coefficients, and the eigenvalues belong to $\mathbb{Q}(\sqrt{5})$ (CP-sensitive).

The CP phase δ_{CP} arises from the interference of these two channels, and its unique source is $\arg(\lambda_3)$ (M52). The value of δ_{CP} itself depends on the VEV dynamics and

belongs to **[P]**, but the uniqueness of the location where CP can enter is closed in layer M.
Lean verification: `M57_both_sigma_fixed` (A5CP.lean).

3.6 Dependency structure between theorems

The eight theorems in this section converge from three independent logical sequences.

Sequence 1 (CP impossibility): M50 → M52b → M52. From the fact that all representations of A_5 are real type (M50), the reality of CG tensors follows (M52b), leading to the uniqueness of the CP source (M52).

Sequence 2 (CP admissibility): M51 → M52a → M52. The complex type of 27 in E_6 (M51) and the uniqueness of cubic invariants (M52a) converge to support M52.

Sequence 3 (McKay–CP structure): M49 (§2.5) → M54 → M53, M55, M56, M57. From the synthesis of the McKay–family dictionary (M49) and Galois involution (M54), the McKay–CP correspondence (M53), the action on 248 (M55), uniqueness (M56), and channel splitting (M57) are derived.

The three sequences merge at M52 and M57, establishing a complete CP structure where "the only source of CP is $\arg(\lambda_3)$" and "mass is CP-sensitive and the mixing angle is CP-even."

4 Uniqueness of $E_8 \to E_6 \times A_2$

In this section, we show that the branch from E_8 to $E_6 \times A_2$ is unique under three conditions: product structure (C1), complex type representation (C2), and family dimension 3 (C3) (M58). Furthermore, we re-derive the same conclusion from an alternative path of node removal in affine $\hat{E}_8$ (M59–M61), confirming that the two paths converge to the same structure.

4.1 M58: Maximal subalgebra selection theorem

The maximal subalgebras of E_8 have been fully classified by Dynkin (1957 [7]). There are a total of eight types: five regular and three special; see Table 7.

Table 7 Maximal subalgebras of E_8.

#	Candidate H	Type	248 branching	dim(adj)
1	$D_8 = \mathrm{SO}(16)$	regular	$120 + 128_s$	120
2	$A_8 = \mathrm{SU}(9)$	regular	$80 + 84 + \overline{84}$	80
3	$A_4 \times A_4 = \mathrm{SU}(5)^2$	regular	$\mathrm{adj}^2 + (5, 10) + \mathrm{c.c.}$	48
4	$E_6 \times A_2$	regular	$78 + 8 + (27, 3) + \mathrm{c.c.}$	86
5	$E_7 \times A_1$	regular	$133 + 3 + (56, 2)$	136
6	$G_2 \times F_4$	special	$14 + 52 + (7, 26)$	66
7	A_1 (special)	special	—	3
8	B_2 (special)	special	—	10

The three selection criteria used in this paper are as follows.

C1 (product structure).

H is the product of at least two simple factors. The condition is determined by algebraic fact [**M**], but the motivation for requiring a product structure—the separation of gauge and family structures—involves a physical judgment [P-leaning]. However, the bottom-up path in §4.2 (M59–M61) reaches the same conclusion independently of C1, so when the two paths are combined, the P-like element of C1 is absorbed. C1 eliminates four candidates: D_8, A_8, A_1(special), and B_2(special).

C2 (complex type representation).

The branch of 248 to H contains a complex component ($R \not\cong \bar{R}$). The condition is determined by computing the Frobenius–Schur indicator [**M**]; the motivation for requiring this is the existence of a CP phase receptacle (§3.2), which references the observational fact that CP violation is physically realized [**E**]. C2 further eliminates $E_7 \times A_1$ and $G_2 \times F_4$.

The exclusion of $E_7 \times A_1$ merits careful consideration. The representation **56** of E_7 is pseudo-real ($\nu_2 = -1$), not complex. $(56, 2)$ lacks compatibility with the Yukawa coupling requiring a Sym^3 structure and is also incompatible with C3 because its family dimension is 2. $G_2 \times F_4$ is excluded because neither G_2 nor F_4 has non-trivial Dynkin diagram automorphisms, so all representations are real type.

C3 (family dimension) [E].

The dimension of the fundamental representation of one of the factors is 3. This is a reference to the observed fact of 3 generations [**E**]. C3 eliminates $A_4 \times A_4$ (family dimension $= 5$).

The results of applying the three conditions are summarized in Table 8.

Table 8 Application of selection criteria C1–C3.

Candidate	C1	C2	C3	Result
D_8	×	—	—	excluded
A_8	×	—	—	excluded
$A_4 \times A_4$	✓	✓	× (5)	excluded
$E_6 \times A_2$	✓	✓	✓ (3)	**survives**
$E_7 \times A_1$	✓	×	× (2)	excluded
$G_2 \times F_4$	✓	×	—	excluded
A_1 (special)	×	—	—	excluded
B_2 (special)	×	—	—	excluded

Theorem 4.1 (M58: Selection of E_8 maximal subalgebras) *Of the 8 maximal subalgebras of E_8, only $E_6 \times A_2$ satisfies C1 (product structure) $\wedge$ C2 (complex type) $\wedge$ C3 (family dimension $= 3$) simultaneously.*

The attribution of M58 is [**M***]. The determination of each condition is algebraically decidable [**M**]. The layer attribution of the motivations for the conditions is: C1 is [P-leaning] (absorbable in M59–M61), C2 is [**E**] (existence of CP violation), and C3 is [**E**] (3 generations). Thus, M58 is a conditional M-layer theorem with [**E**] as external input.
Lean verification: `M58_E6A2_unique_selection`, `M58_iff` (E8MaxSub.lean).

4.2 M59: Complete classification of $\hat{E}_8$ single-node removal

M58 was a top-down selection from the algebraic structure of the maximal subalgebras of E_8. In this section, we arrive at the same conclusion through a bottom-up path of node removal in the affine $\hat{E}_8$ McKay graph.

Theorem 4.2 (M59) *When node k ($k = 0, \ldots, 8$) is removed from affine $\hat{E}_8$, the remaining Dynkin type is as given in Table 9.*

Table 9 Complete classification of $\hat{E}_8$ single-node removal.

Node	Mark	$2I$ irrep	Residual Dynkin type	Lie group
0	1	ρ_1 (affine)	E_8	E_8
1	2	spinorial	D_8	SO(16)
2	3	ρ_3	A_8	SU(9)
3	4	ρ_4	$A_1 + A_7$	SU(2) × SU(8)
4	6	spinorial	$A_1 + A_2 + A_5$	SU(2) × SU(3) × SU(6)
5	5	ρ_5	$A_4 + A_4$	$\mathrm{SU}(5)^2$
6	4	spinorial	$A_3 + D_5$	SU(4) × SO(10)
7	3	ρ_3'	$E_6 + A_2$	E_6 × SU(3)
8	2	spinorial	$A_1 + E_7$	SU(2) × E_7

The proof is performed by finite enumeration (9 possibilities). The connected components of the remaining subgraphs after removing each node are determined, and the Dynkin type is identified by arm-length analysis. In all 9 cases, the total rank of the residual type is $8 = \mathrm{rank}(E_8)$.

In Table 9, removal of node 0 is the affine node removal, recovering finite type E_8. Only removal of node 4 (branch point, mark = 6) results in a 3-component residual.
Lean verification: `nodeRemovalResult`, `M59_rank_check`, `three_components_unique` (E8BlowUp.lean, `decide`).

4.3 M60: Uniqueness of E_6 factor

Theorem 4.3 (M60) *Of the nine possibilities in M59, only removal of node 7 results in a residual Dynkin type containing an E_6 factor.*

This is verified by scanning Table 9.

Lean verification: `M60_E6_factor_unique` (E8BlowUp.lean, `decide`).

4.4 M61: Mark-3 dichotomy

Theorem 4.4 (M61: Mark-3 dichotomy) *In $\hat{E}_8$, the only nodes with* mark $= 3$ *are* $\{2, 7\}$. *Both are A_5-type (§2.3) and correspond to the Galois pair $\{\rho_3, \rho_3'\}$ (§2.4). However, the removal results are fundamentally different:*

Node	*A_5 irrep*	*Removal result*	*Product?*	*E_6 factor?*
7	ρ_3'	$E_6 + A_2$	✓	✓
2	ρ_3	A_8	×	×

Removing node 7 yields $E_6 \times A_2$, opening a path to trinification. Removing node 2 yields $A_8 = \mathrm{SU}(9)$ (single factor), making gauge/family separation impossible.

In the Galois pair, the asymmetry whereby ρ_3' (node 7) carries the family index and ρ_3 (node 2) remains inside E_6 follows deterministically from the topology of the McKay graph. Node 7 is on the side of the main chain closer to the affine node 0 (connected via node 8), while node 2 is directly connected to the branch point (node 4). This difference fundamentally alters the connectivity structure of the graph after removal.

M61 can be restated as follows: the only node removal that simultaneously satisfies "E_6 factor $\wedge$ A_2 factor $\wedge$ mark $= 3$" is node 7. This is a node-removal version of M58 (C1 + C2 + C3).

Lean verification: `M61_mark3_dichotomy`, `galois_pair_asymmetry`, `galois_separation` (E8BlowUp.lean, `decide`).

4.5 Convergence of dual paths

The theorem system in this section establishes the uniqueness of $E_8 \to E_6 \times A_2$ through two independent paths: top-down (M58: Dynkin classification + C1/C2/C3 selection) and bottom-up (M59–M61: complete enumeration of McKay graph nodes). The triple fixation, combining these with M49 (parity classification + Galois pairs), is summarized in Table 10.

Table 10 Triple convergence of independent paths.

Route	Starting point	Method	Conclusion
M49	McKay parity + Galois pair	Character analysis	$(27, 3)$: $3 = \rho_3'$
M58	Dynkin maximal sub-classification	C1+C2+C3	$E_6 \times A_2$ unique
M59–M61	$\hat{E}_8$ node removal	Complete enumeration	Node 7 unique

The fact that three routes converge to the same structure from different starting points and proof methods supports the structural necessity of $E_6 \times A_2$.

4.6 E_6 internal node removal and connection to trinification

Sections 4.1–4.5 established the uniqueness of $E_8 \to E_6 \times A_2$. Before discussing $E_6 \to \mathrm{SU}(3)^3$ in §5, we analyze node removal within E_6 to confirm why trinification cannot be achieved by simple Dynkin diagram operations.

The Dynkin diagram of E_6 has the structure:

$$6—5—4—3—1, \qquad 4—2 \quad \text{(branch)}. \tag{8}$$

Removing the branch point (node 4, mark = 6) yields $A_2 + A_2 + A_1 = \mathrm{SU}(3) \times \mathrm{SU}(3) \times \mathrm{SU}(2)$, which is the node removal within E_6 that generates the most factors. However, this does not reach the trinification $\mathrm{SU}(3)^3$.

$E_6 \to \mathrm{SU}(3)^3$ is *not* node removal in the Dynkin diagram but a regular subalgebra embedding. The operation of dividing the root system of E_6 into three A_2 components is a structure that goes beyond the combinatorial theory of nodes and will be discussed in §5 as P4 (trinification).

Lean verification: `e4_removal_three_components`, `two_step_trinification_chain` (E8BlowUp.lean).

5 Trinification: $E_6 \to \mathrm{SU}(3)^3$

The purpose of this section is to clarify what is already mathematically fixed and what remains as a physical judgment in the process of moving from E_6 to the trinification $\mathrm{SU}(3)^3$ (Glashow 1984 [8]). To anticipate the conclusion, the maximal subalgebra candidate space of E_6 itself is finitely fixed by the Dynkin classification, and furthermore, $\mathrm{SU}(3)^3$ is conditionally unique under two independent conditions by M62 and M63. Therefore, the substance of P remaining in the trinification stage is limited to two points: the physical principle that actually adopts $\mathrm{SU}(3)^3$ and the identification of which of its three factors is read as color.

5.1 P4a: Fixing the candidate space

The maximal subalgebras of E_6 have been fully classified by Dynkin (1957 [7]). There are a total of six types: three regular and three special; see Table 11.

Table 11 Maximal subalgebras of E_6.

#	Candidate H	Dynkin type	Factors	27 branching	dim(adj)
R1	$\mathrm{SO}(10) \times \mathrm{U}(1)$	D_5 + ab.	1 + ab.	$16 \oplus 10 \oplus 1$	46
R2	$\mathrm{SU}(6) \times \mathrm{SU}(2)$	$A_5 \times A_1$	2 (non-iso.)	$(6,2) \oplus (15,1)$	38
R3	$\mathrm{SU}(3)^3$	A_2^3	3 (isomorphic)	$(3,\bar{3},1) \oplus (1,3,\bar{3}) \oplus (\bar{3},1,3)$	24
S1	$\mathrm{SU}(3) \times G_2$	$A_2 \times G_2$	2 (non-iso.)	$(3,7) \oplus \cdots$	22
S2	$\mathrm{Sp}(8)$	C_4	1	27 (irreducible)	36
S3	F_4	F_4	1	$26 \oplus 1$	52

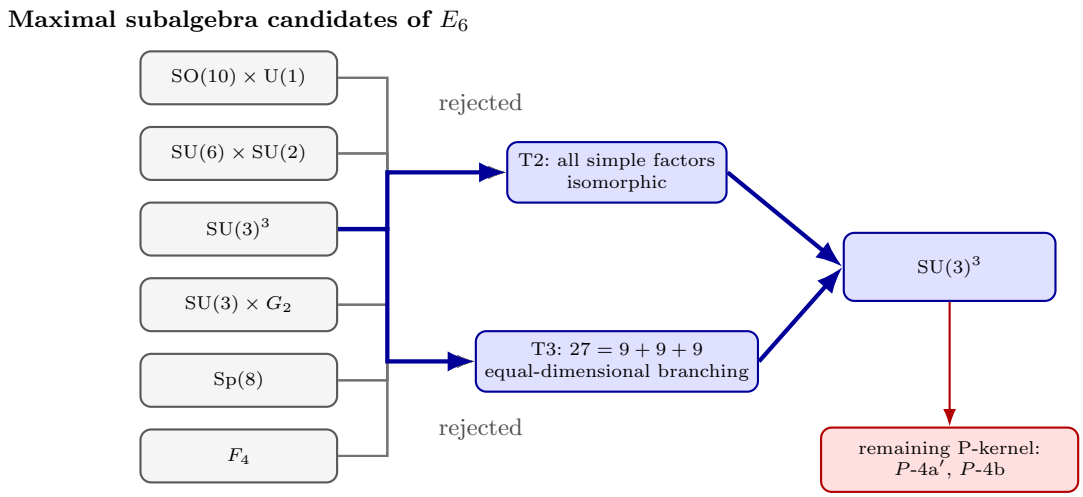

Fig. 4 Candidate-space filtering for trinification inside E_6. Among the six maximal subalgebra candidates, the independent criteria T2 (all simple factors isomorphic) and T3 (equal-dimensional branching of the 27) both select $\mathrm{SU}(3)^3$ uniquely. The residual physical arbitrariness is localized at P-4a$'$ and P-4b.

It has been verified in Lean that the sum of the dimensions of the 27 branching across all 6 candidates equals 27.

Two independent uniqueness theorems hold on this candidate space.

Theorem 5.1 (M62: T2 selection) *The only maximal subalgebra of E_6 that has two or more non-abelian simple factors and whose factors are all isomorphic is* $\mathrm{SU}(3)^3$.

The T2 condition ("all factors isomorphic") is a well-defined predicate determined by scanning 6 candidates. Among the multifactor types, $\mathrm{SU}(6) \times \mathrm{SU}(2)$ has $\mathrm{SU}(6) \neq \mathrm{SU}(2)$, and $\mathrm{SU}(3) \times G_2$ has $\mathrm{SU}(3) \neq G_2$, so they are incompatible. Single-factor types ($\mathrm{SO}(10) \times \mathrm{U}(1)$, $\mathrm{Sp}(8)$, F_4) do not satisfy the prerequisite (≥ 2 factors).

Theorem 5.2 (M63: T3 selection) *The only maximal subalgebra of E_6 whose 27 branching decomposes into two or more equal-dimensional components is* $\mathrm{SU}(3)^3$.

The T3 condition requires $27 = 9 + 9 + 9$. The decompositions $16 + 10 + 1$ for $\mathrm{SO}(10) \times \mathrm{U}(1)$, $15 + 12$ for $\mathrm{SU}(6) \times \mathrm{SU}(2)$, and $26 + 1$ for F_4 are not equal-dimensional. In $\mathrm{Sp}(8)$, 27 is irreducible.

T2 and T3 are logically independent conditions, but they select the same candidate $\mathrm{SU}(3)^3$. This overdetermination has been independently verified in Lean (`T2_alone_sufficient` and `T3_alone_sufficient`).

The convergence of two independent conditions to the same solution reflects a structural singularity of $\mathrm{SU}(3)^3$.

Figure 4 summarizes the candidate-space filtering behind M62/M63.

The content of P4a belongs entirely to the M layer. Dynkin's classification, the structural properties of each candidate, and the uniqueness under T2/T3 are all closed by finite enumeration + `decide`.
Lean verification: `M62_equal_factor_selection`, `M63_equal_dim_branching`, `three_factors_unique` (E6Trinification.lean).

5.2 Supplemental consistency check

The conditional uniqueness of $\mathrm{SU}(3)^3$ is established by M62/M63. In this section, a three-axis consistency check based on M-layer theorems is recorded as a supplement. The scoring is not a substitute for the uniqueness theorem in §5.1, but rather an auxiliary means of confirming the physical differences among the six candidates.

Each axis is based on existing M-layer theorems. Family consistency (axis I) evaluates the correspondence between the outer $(27, 3)$ family-triplet structure (M49) and the inner 27-branching structure. CP consistency (axis II) evaluates whether the unique CP source $\arg(\lambda_3)$ (M52) is cyclically distributed on the d_{ijk} tensor blocks. Democracy consistency (axis III) corresponds to the degree to which the T2 condition of M62 holds. The results are shown in Table 12.

Table 12 Three-axis consistency scoring for E_6 maximal subalgebras.

Candidate H	Axis I	Axis II	Axis III	Total
$\mathrm{SU}(3)^3$	3	3	3	9/9
$\mathrm{SO}(10) \times \mathrm{U}(1)$	2	1	1	4/9
$\mathrm{SU}(6) \times \mathrm{SU}(2)$	1	1	1	3/9
$\mathrm{SU}(3) \times G_2$	0	0	1	1/9
$\mathrm{Sp}(8)$	0	0	0	0/9
F_4	0	0	0	0/9

$\mathrm{SU}(3)^3$ is the only candidate that scores the maximum on all three axes, while the other five score 0 or 1 on at least one axis. This structure is robust in the sense of Pareto optimality; $\mathrm{SU}(3)^3$ is no worse than any other candidate on all axes and strictly better on at least one.

This robustness limits the P-layer content of P-4a′. P-4a′ does not assert the conclusion "$\mathrm{SU}(3)^3$ is chosen" (which is closed in M as a conditional consequence of M62/M63), but merely the judgment "equal-rank democracy (T2 condition) is adopted as a physical principle."

5.3 Z_3 cyclic symmetry

$\mathrm{SU}(3)^3$ has a cyclic permutation Z_3 between its three factors:

$$\mathrm{SU}(3)_1 \to \mathrm{SU}(3)_2 \to \mathrm{SU}(3)_3 \to \mathrm{SU}(3)_1. \tag{9}$$

This Z_3 is realized as part of the outer automorphism group of E_6. The branching $(3, \bar{3}, 1) \oplus (1, 3, \bar{3}) \oplus (\bar{3}, 1, 3)$ is cyclically permuted under Z_3. The combined total rank of $\mathrm{SU}(3)^3$ and $\mathrm{SU}(3)_{\mathrm{family}}$ from M58 is $3 \times 2 + 2 = 8 = \mathrm{rank}(E_8)$, consuming the rank of E_8 without excess.
Lean verification: `z3_is_cyclic`, `z3_has_no_fixed_point`, `full_chain_four_SU3` (E6Trinification.lean).

5.4 M/P attribution and the P-nucleus remaining in trinification

The core of the structural P remaining in trinification is compressed into two points: P-4a′ and P-4b. The content of P-4a′ is refined from the vague assertion "choosing trinification" to the limited and falsifiable assertion "adopting the physical principle of equal-rank democracy." P-4b remains as an inevitable physical choice: which of the three cyclically symmetric SU(3) factors under Z_3 is identified as color.

The fact that P-4a′ and P-4b are fundamentally impossible to promote to M is supported by the following proposition.

Proposition 5.3 (M-promotion impossibility) All six maximal subalgebras of E_6 exist as well-defined subalgebras. Based solely on their algebraic structure, there is no mathematical reason to consider any one of them as distinguished from the other five.

Proof By Dynkin's classification theorem, the six candidates correspond to different embeddings of E_6 into the root system, and all are mathematically valid. The T2/T3 condition is a well-defined predicate on the candidate space, but the decision to adopt that predicate as a physical selection criterion is not contained within the algebraic structure. Similarly, the three factors of $\mathrm{SU}(3)^3$ are mathematically equivalent by Z_3 cyclic symmetry, and there is no algebraic reason to distinguish one as color. Therefore, P-4a′ and P-4b structurally remain in [**P**]. □

6 Minimal hypothesis set and epistemological significance

6.1 Determining the minimum set of hypotheses

We systematically examine all 25 P-layer hypotheses accumulated through the research program (complete list in Appendix C) and identify the inevitable hypotheses remaining after pushing M-promotion to its limits. The minimal hypothesis set $\Sigma_{\min}$ consists of six hypotheses:

$$\boxed{\Sigma_{\min} = \{E_{1a},\ E_{1b},\ E_2,\ E_3,\ \text{P-4a}',\ \text{P-4b}\}.} \tag{10}$$

These six hypotheses divide into two groups by their role in the pathway.

Table 13 Core pathway hypotheses.

Hypothesis	Layer	Content	Role
E_{1a}	**[E]**	Spatial dimension $n = 3$	Step 0–1: SO(3) $\to A_5$
E_{1b}	**[E]**	$N_{\text{gen}} = 3$	Step 4–5: C3 for M58
P-4a′	**[P]**	Equal-rank democracy	Step 7: $E_6 \to \text{SU}(3)^3$

Core pathway hypotheses *(*$A_5 \to E_6 \times A_2 \to \text{SU}(3)^3$*):*
SM identification hypotheses *(*$\text{SU}(3)^3 \to$ ***SM):***

Table 14 SM identification hypotheses.

Hypothesis	Layer	Content	Role
E_2	**[E]**	Strong force is non-abelian	Step 8: SM identification
E_3	**[E]**	Asymptotic freedom	Step 8: SM identification
P-4b	**[P]**	Color identification	Step 8: $\text{SU}(3)^3 \to$ SM

6.2 Demonstration of completeness

By combining the six hypotheses of Σ_{min} with the M-layer theorems, all steps in the path are uniquely determined; see Table 15.

Table 15 Completeness of the pathway.

Step	Input	M-layer theorem	Output
$0 \to 1$	E_{1a} $(n = 3)$	Klein [9]	Finite subgroups of SO(3)
$1 \to 2$	E_{1a} + Born law [4]	M01–M29	Uniqueness of A_5
$2 \to 3$	—	Schur multiplier	$2I$ (double cover)
$3 \to 4$	—	McKay [1]	$\hat{E}_8$
$4 \to 5$	E_{1b} $(N_{\text{gen}} = 3)$	M58	$E_6 \times A_2$ unique
$5 \to 6$	—	M49–M57	Family $= \rho_3'$, CP structure
$6 \to 7$	P-4a′	M62/M63	$\text{SU}(3)^3$ unique
$7 \to 8$	P-4b + E_2 + E_3	SM identification	SM

The P-layer hypothesis is not used in Steps 2–4 and 5–6.

The "Born law" in Step $1 \to 2$ refers to the main theorem of Paper I [4]. Paper I shows that when the Born probability law $P(i) = n_i^2/|G|$ (where n_i is the dimension of the irreducible representation) is defined on a finite subgroup G of SO(3), the group G must be non-solvable for this distribution to differ from the uniform distribution. Since the only non-solvable finite subgroup of SO(3) under $n = 3$ is A_5 (Klein [9]), A_5 can be uniquely derived from the non-triviality of the Born law and $n = 3$. This process is verified as M01–M29 in Lean.

6.3 Demonstration of irreducibility

We examine the consequences of removing any one hypothesis from $\Sigma_{\min}$ and show that the unique determination of the path is broken in each case.

If E_{1a} is removed: SO(3) is not determined. For $n \neq 3$, the Klein classification is qualitatively different (no non-solvable group for $n = 2$, no exceptional finite subgroup corresponding to A_5 for $n \geq 5$). The entire path collapses.

If E_{1b} is removed: The C3 condition (family dimension = 3) for M58 becomes undefined. For $N_{\text{gen}} \neq 3$, the E_8 maximal subalgebra selection changes ($N_{\text{gen}} = 2$ leaves $E_7 \times A_1$ as a candidate; $N_{\text{gen}} = 5$ makes $A_4 \times A_4$ a candidate). The uniqueness of $E_6 \times A_2$ is broken.

If E_2 is removed: The three factors of $\text{SU}(3)^3$ are algebraically equivalent under Z_3 cyclic symmetry, but in SM, one exhibits confinement as color. This asymmetric physical role cannot be determined without E_2.

If E_3 is removed: Asymptotic freedom is independent of non-abelianity (E_2) and is an empirical fact depending on the actual number of flavors. Removing E_3 makes it impossible to verify the consistency of coupling running in the trinification $\to$ SM connection.

If P-4a′ *is removed:* It becomes impossible to uniquely select $\text{SU}(3)^3$ from the 6 maximal subalgebras of E_6. At least 3 candidates remain: $\text{SO}(10) \times \text{U}(1)$, $\text{SU}(6) \times \text{SU}(2)$, and $\text{SU}(3)^3$.

If P-4b *is removed:* Even after $\text{SU}(3)^3$ is selected, the physical roles of the three factors remain undetermined. Due to Z_3 cyclic symmetry, it is not possible to determine which factor is $\text{SU}(3)_C$ from algebraic means.

In all cases, path uniqueness is broken; therefore irreducibility is satisfied.

6.4 Previously M-promoted hypotheses

As the research program progressed, several hypotheses initially belonging to layer P were promoted to layer M. Since these have been established as theorems, they are no longer hypotheses and are not included in $\Sigma_{\min}$.

Table 16 P-layer hypotheses promoted to M-layer.

Old P hypothesis	M-layer theorem	Key to promotion
P-2b (A_5 uniqueness)	Klein [9] + M01–M29	$n = 3$ + non-solvability
P-3c ($E_8 \to E_6 \times A_2$)	M58 (§4.1)	C1+C2+C3 finite enumeration
P-3d ($\text{SU}(3)_{\text{fam}} = \rho_3'$)	M49 (§2.5)	Parity + node 7 removal
P-5b ($\text{CP} = \arg(\lambda_3)$)	M52 (§3.3)	Cubic invariant uniqueness

6.5 Distribution and structural significance of P

The distribution of the six $\Sigma_{\min}$ hypotheses along the path shows a clear pattern:

$$E_{1a} \longrightarrow \underbrace{\text{Steps 1–4}}_{\text{M only}} \longrightarrow E_{1b} \longrightarrow \underbrace{\text{Steps 5–6}}_{\text{M}} \longrightarrow \text{P-4a}' \longrightarrow \underbrace{\text{P-4b} + E_2 + E_3}_{\text{SM identification}}. \tag{11}$$

The core pathway hypotheses (E_{1a}, E_{1b}, P-4a$'$) are distributed at entry, intermediate, and exit points of the pathway, while the SM identification hypotheses (E_2, E_3, P-4b) are concentrated in the final step. Crucially, the core interval—Steps 1–4 ($A_5 \to 2I \to \hat{E}_8$) and Steps 5–6 (family/CP structure)—is complete in the M layer without P-layer intervention.

Note that $\Sigma_{\min}$ is a list of hypotheses *within* the path and does not include the framework selection FA_1 (§2.1). FA_1 is not an individual bridge hypothesis but a framework hypothesis for the entire program, consisting of two components: FA_{1a} (canonical lift) and FA_{1b} (physical adoption); see §2.1 for the detailed decomposition, the non-arbitrariness argument, and the counterfactual analysis (Table 1). Therefore, the overall assumption structure of this paper is:

$$\underbrace{\mathrm{FA}_1 = \mathrm{FA}_{1a} + \mathrm{FA}_{1b}}_{\text{Framework}} + \underbrace{\Sigma_{\min} = \{E_{1a}, E_{1b}, E_2, E_3, \text{P-4a}', \text{P-4b}\}}_{\text{6 hypotheses within the path}}. \tag{12}$$

By explicitly stating this structure—and by decomposing FA_1 into a mathematically motivated component (FA_{1a}) and a genuine physical commitment (FA_{1b})—we prevent the criticism of "not counting the heaviest assumptions."

6.6 Transparency of hypotheses

The methodological feature of this program lies not in the total number of assumptions, but in the consistent traceability of which claims are mathematical consequences, which are physical hypotheses, and which are citations of observed facts. The important point is not to compete on the number of assumptions, but to clearly indicate where arbitrariness remains in the theory.

6.7 Falsifiability of P-4a$'$ and P-4b

In this paper, only P-4a$'$ and P-4b remain in the P layer. Future observations and model comparisons will focus on these two hypotheses.

If P-4a$'$ is rejected, the conditional uniqueness under M62/M63 itself is not broken; rather, the reason for physically adopting $\mathrm{SU}(3)^3$ is lost. Other E_6 maximal subalgebra candidates such as $\mathrm{SO}(10)\times\mathrm{U}(1)$ and $\mathrm{SU}(6)\times\mathrm{SU}(2)$ would be reintroduced as physical candidates.

Similarly, if P-4b is negated, it is not that the algebraic structure $\mathrm{SU}(3)^3$ itself is broken, but rather that the assignment of physical roles to its three factors was incorrect. Phenomena specific to trinification, such as decay modes sensitive to color assignment and exotic states, serve as direct tests.

Table 17 Proton decay branching-ratio discriminants. $R_{\pi/K}$ is defined in Eq. (13). The $\mathrm{SU}(3)^3$ column follows from Z_3 cyclic symmetry (M62, Lean verified) and is independent of M_{tri}. Hadronic matrix elements carry $\sim 20\%$ lattice uncertainty.

Observable	SU(5)	SO(10) (SUSY)	$\mathrm{SU}(3)^3 + A_5$
Dominant mode	$p \to e^+\pi^0$	$p \to \bar{\nu}K^+$	$p \to e^+\pi^0$
$R_{\pi/K}$	≈ 43	$\ll 1$	1–5
$R_{\ell/\nu}$	> 1	< 1	≈ 1
Flavor origin	CKM	CKM + seesaw	A_5 CG (ρ_3')
M_{GUT}-indep.?	no	no	yes (ratio)
P-4b test	—	—	✓

The most robust quantitative test is the proton decay branching-ratio pattern. The Z_3 cyclic symmetry of $\mathrm{SU}(3)^3$ (M62, Lean verified: `z3_is_cyclic`, `z3_has_no_fixed_point`) equalizes the three inter-sector gauge couplings at the trinification scale M_{tri}. In dimension-6 gauge-boson-mediated decay, the ratio

$$R_{\pi/K} \;=\; \frac{\Gamma(p \to e^+\pi^0)}{\Gamma(p \to \bar{\nu}\, K^+)} \tag{13}$$

factorizes as $R_{\pi/K} = (\text{flavor ratio}) \times (W_\pi/W_K)^2 \times (\text{phase space ratio})$. In minimal SU(5), the flavor factor is $|V_{ud}/V_{us}|^2 \approx 19$ (CKM suppression of the $\Delta S = 1$ channel), giving $R_{\pi/K} \approx 43$. In $\mathrm{SU}(3)^3$ trinification, Z_3 democracy sets the flavor ratio to unity at M_{tri}; Yukawa running corrections are negligible ($\delta f/f \sim y_s^2 \ln(M_{\mathrm{tri}}/m_p)/(16\pi^2) \ll 1$). Including $\sim 20\%$ lattice uncertainty on the hadronic matrix elements $W_{\pi,K}$, one obtains

$$R_{\pi/K}^{\mathrm{SU}(3)^3} \;\in\; [1,5], \qquad R_{\pi/K}^{\mathrm{SU}(5)} \approx 43. \tag{14}$$

The factor-of-~ 19 separation is structural: it originates entirely from CKM vs. Z_3-democracy and does *not* depend on M_{tri} (which cancels in the ratio) or on threshold corrections. Table 17 summarizes the comparison.

The ratio $R_{\ell/\nu} = \Gamma(p \to \ell^+ + \text{meson})/\Gamma(p \to \bar{\nu} + \text{meson})$ provides a second discriminant. The Z_3 symmetry relates the $(Q \to L)$ and $(Q \to \bar{Q})$ inter-sector transitions symmetrically, yielding $R_{\ell/\nu} \approx 1$, in contrast to SUSY SO(10) where dimension-5 operators give $R_{\ell/\nu} < 1$.

For P-4b, different assignments of $\mathrm{SU}(3)_C$ among the three Z_3-equivalent factors lead to different branching-ratio patterns, making P-4b an experimentally falsifiable hypothesis. These predictions can be tested at Hyper-Kamiokande and DUNE.

7 Comparison with other theories and future prospects

7.1 Relationship with existing discrete symmetry models

Numerous models based on discrete flavor symmetry have been proposed (see reviews [10, 11]). A_4 (order 12) is widely used to explain the tri-bimaximal mixing

pattern [12, 13], and S_4 (order 24) has been applied to atmospheric mixing angle maximality [14]. $\Delta(27)$ (order 27) contains a complex structure for CP violation [15].

The A_5 (order 60) adopted by this program is fundamentally different in the following respects.

First, A_5 is uniquely characterized as a non-solvable finite subgroup of SO(3) (Paper I [4] main theorem). A_4 and S_4 are both solvable and lack a comparable uniqueness justification.

Second, the fact that all irreducible representations of A_5 are real type (M50) allows a clear attribution of the CP phase origin to structures outside A_5. For A_4 and $\Delta(27)$, complex representations exist, and the origin of CP is intertwined with the internal structure of the group, making M/P separation difficult.

Third, the connection to E_8 through the McKay correspondence is unique to A_5. A_4, S_4, and $\Delta(27)$ are finite subgroups of SO(3), but the McKay correspondence holds for finite subgroups of SU(2)—that is, double covering groups. The double cover $2I$ of A_5 gives an E_8-type McKay graph, but the double covers $2T$ of A_4 (E_6 type) and $2O$ of S_4 (E_7 type) do not reach E_8.

On the other hand, this program has significant limitations that need explicit statement. First, individual fermion masses and mixing angles are not yet predicted; these require completion of the VEV analysis for the 19-parameter A_5-invariant potential, which is ongoing. However, the framework does yield structural predictions at the present stage: the proton decay ratio $R_{\pi/K} \in [1, 5]$, qualitatively distinct from $R^{\mathrm{SU}(5)}_{\pi/K} \approx 43$ (§6.7, Table 17), and the structural necessity of $\theta_{13} \neq 0$. On the latter point, A_4 models predicted $\theta_{13} = 0$ under tri-bimaximal mixing and were falsified by Daya Bay (2012); in the A_5 framework, ρ_3 and ρ_3' are non-isomorphic Galois conjugates (M29), and the Channel III mixing matrix ($\rho_3 \otimes \rho_3' \otimes \rho_4$, M57) generically has a non-vanishing 1–3 component, making $\theta_{13} \neq 0$ a structural consequence rather than a parameter adjustment. Second, the rationale for physically adopting the McKay path (FA_1) is a framework choice fundamentally different from "justification by fitting."

7.2 Relationship with E_8 unified theory

The approach of using E_8 as a unifying group has been widely studied in the context of heterotic string theory [6]. In this program, E_8 is obtained independently of string compactification, from the McKay correspondence. The idea of applying the algebraic structure of E_8 to family unification is also seen in Ramond [16], but this program starts from the McKay correspondence.

The Griess–Ryba [17] classification of finite group embeddings into E_8 refines the positioning of $2I$ within E_8. The G1′ programme of this program aims to concretize the connection between McKay and Lie algebraic paths by computing the structure of $C_{E_8}(2I)$ using GAP.

Dechant's [18] Clifford algebraic approach illuminates the icosahedron–E_8 relationship from a different perspective. Its relation to this program is formally recorded in G1SelCore.lean as `dechant_not_SU3_class`.

7.3 Future outlook

Short term.

(i) Paper V (CP violation): based on the CP structure of M50–M57 (§3), quantitative prediction of δ_{CP} via swap-odd structures and VEV mechanics. The scope of M52 is limited to renormalizable theory; a systematic analysis of higher-dimensional effective operators is also included. (ii) Numerical refinement of $R_{\pi/K}$: the structural prediction $R_{\pi/K} \in [1, 5]$ (§6.7) will be sharpened by incorporating one-loop RG running from M_{tri} to m_p and the A_5 Clebsch–Gordan corrections from the ρ_3' family structure (M49). A preliminary analysis using one-loop SM running gives $\alpha_2^{-1}(M_{\mathrm{tri}}) = \alpha_3^{-1}(M_{\mathrm{tri}})$ at $\log_{10}(M_{\mathrm{tri}}/\mathrm{GeV}) \approx 17.0$, with threshold corrections admitting a window $\log_{10}(M_{\mathrm{tri}}/\mathrm{GeV}) \in [15, 19]$. (iii) A_5-specific corrections to $R_{e/\mu} = \Gamma(p \to e^+\pi^0)/\Gamma(p \to \mu^+\pi^0)$, arising from the ρ_3' tensor product decomposition $\rho_3' \otimes \rho_3' = \rho_1 \oplus \rho_3 \oplus \rho_5$ (M01, Lean verified), will be computed after Phase 3 (Channel III Clebsch–Gordan completion).

Mid-term.

(iv) G1′ programme: computing $C_{E_8}(2I)$ in GAP to directly connect the McKay path ($A_5 \to 2I \to E_8$) and the Dynkin–Lie path ($E_8 \to E_6 \times A_2 \to \mathrm{SU}(3)^3$), potentially elevating "uniqueness as selection" (M58) to "uniqueness as embedding." (v) Gauge coupling unification: extending the preliminary one-loop analysis of (ii) to two-loop running with threshold corrections, and comparing the $\mathrm{SU}(3)^3$ unification window with those of $\mathrm{SO}(10) \times \mathrm{U}(1)$ and $\mathrm{SU}(6) \times \mathrm{SU}(2)$ to further constrain P-4a′.

Long term.

(vi) Origin of the Lorentz group and gravity: deriving spacetime symmetries from the same algebraic framework; null-count kernels and the edge-strain variational principle are candidate approaches. (vii) Complete description of the mass hierarchy: the A_5 flavor structure gives a mass/mixing Galois splitting (M57), but specific fermion masses depend on VEV dynamics. Phase structure analysis of the 19-parameter A_5-invariant potential is underway.

8 Conclusion

In this paper, the entire pathway $A_5 \to 2I \to \hat{E}_8 \to E_6 \times A_2 \to \mathrm{SU}(3)^3 \to \mathrm{SM}$ was dissected under epistemological three-layer separation (M/P/E), and the following two main results were established.

Main result 1 (rigidity).

Under the framework hypothesis FA_1, the core pathway $A_5 \to 2I \to \hat{E}_8 \to E_6 \times A_2$ is uniquely fixed in layer M with E_{1a} ($n = 3$) and E_{1b} ($N_{\mathrm{gen}} = 3$) as the sole external inputs. More than 60 named theorems (M01–M63 and supporting results), including M49–M63, have been formally verified by Lean 4 with `sorry = 0` and `axiom = 0`; the precise scope of what the type-checker guarantees is detailed in §A.2. Note that M52

(uniqueness of CP source) holds only for the renormalizable sector; additional CP sources from effective operators of dim ≥ 5 fall within the scope of Paper V.

Main result 2 (localization).

$\Sigma_{\min} = \{E_{1a}, E_{1b}, E_2, E_3, \text{P-4a}', \text{P-4b}\}$ is complete and irreducible. The structural degrees of freedom belonging to the P-layer are localized at two points: P-4a′ and P-4b.

The epistemological significance of these results can be summarized as follows.

First, this paper establishes the precise identification of the M/P boundary. This boundary accurately identifies areas that need improvement—for example, the M-promotion of P-4a′ or the direct McKay–Lie pathway connection by the G1′ programme.

Second, all M-layer theorems have been mechanically verified using a Lean 4 type checker, and these mathematical consequences are irrevocable. Only the two P-layer hypotheses (P-4a′ and P-4b) and the physical-adoption component FA_{1b} of the framework hypothesis may be modified by future observations or theoretical developments. If FA_1 is rejected, the M-layer theorems remain valid as pure mathematics; only their physical interpretation is withdrawn (§2.1, Table 1).

Third, the concentration of falsifiability on P-4a′ and P-4b limits the focus of experimental verification (§6.7). The Z_3 cyclic symmetry of $\text{SU}(3)^3$ (M62) yields a concrete, M_{tri}-independent discriminant: $R_{\pi/K} \in [1, 5]$, separated by a factor of ~ 19 from the minimal SU(5) prediction (Table 17). If experiments reject P-4a′, it is not the rigidity of the core pathway that collapses, but only the physical adoption of trinification, and the candidate space of E_6 (M62/M63) is reopened.

Finally, the methodology of this paper—attributing all claims to M/P/E and machine-verifying the M layer—is not specific to the A_5 program but is a general framework applicable to any BSM physics theory. Maintaining constant traceability of the boundary between mathematical consequences and physical hypotheses, and supporting this with formal verification, proposes a new standard for assumption management in theoretical physics.

Acknowledgments. The formal verification in this paper is based on the Lean 4 theorem prover [19] and the Mathlib mathematical library [20]. The author thanks the developer communities of both projects.

Statements and Declarations.

Funding.

This research received no external funding.

Competing interests.

The author declares no competing interests.

Ethics approval and consent to participate.

Not applicable.

Consent for publication.

Not applicable.

Data availability.

No experimental data were generated or analysed in this study. All numerical values of physical constants are taken from published sources cited in the bibliography. The Lean 4 source code and numerical computations supporting the results are publicly available at https://github.com/nuu/A5_SM_Pathway and permanently archived at Zenodo (https://doi.org/10.5281/zenodo.19378068) under the Apache License 2.0.

Materials availability.

Not applicable.

Code availability.

The complete Lean 4 source code for the formal verification (sorry = 0, axiom = 0) and the theorem–identifier correspondence table are publicly available at https://github.com/nuu/A5_SM_Pathway and permanently archived at Zenodo (https://doi.org/10.5281/zenodo.19378068) under the Apache License 2.0. The repository includes all files necessary to reproduce the verification: `lakefile.lean`, Mathlib dependency specifications, and per-section Lean source files (`Section1_Introduction.lean` through `Section8_Conclusion.lean` and auxiliary modules).

Author contribution.

M. Numagaki is the sole author and is responsible for the conception of the theoretical framework, all mathematical proofs and their Lean 4 formalization, the numerical analysis, and the writing of the manuscript.

Use of AI-assisted tools.

Use of Generative AI and AI-assisted technologies in the writing process During the preparation of this work the author used ChatGPT in order to improve the readability and language of the manuscript. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Appendix A Lean 4 verification system

A.1 Project overview

The M-layer theorems in this research program are implemented as a formal proof project called "A5_SM_Pathway" using Lean 4 + Mathlib. While there are precedents for large-scale formal verification in mathematics (such as the Kepler conjecture [21]), to the best of the author's knowledge, there are no precedents for applying formal verification of this scale to the theoretical structure of elementary particle physics.

Table A1 A5_SM_Pathway project statistics.

Item	Value
Project name	A5_SM_Pathway
Number of files	∼45
Total lines	∼10 800
`sorry`	0
`axiom`	0
Main theorems	60+ (named M01–M63)
Total Lean `theorem` declarations	∼200 (incl. supporting lemmas)
Build system	`lake build`
Zenodo DOI	10.5281/zenodo.19378068
GitHub	github.com/nuu/A5_SM_Pathway

`sorry = 0` means all theorems have been proven with no unproven holes. `axiom = 0` means no additional axioms beyond the standard Lean/Mathlib axioms have been used.

A.2 Scope and limitations of formal verification

It is essential to state precisely what the Lean type-checker does and does not guarantee. Table A2 summarizes the boundary.

Table A2 Scope of the Lean 4 verification.

Verified by the type-checker	**Not verified by the type-checker**
Given the hand-encoded finite data (character tables, Dynkin diagram adjacencies, Cartan matrix entries, maximal-subalgebra lists), all logical consequences drawn from those data are machine-checked with no gaps.	The correctness of the hand-encoded data itself. For example, the values $\chi_{\rho_3}(C_5^+) = \varphi = (1+\sqrt{5})/2$ and `nodeRemovalResult .n7` = [`.E6`, `.A 2`] are typed by the author. If an entry were mis-transcribed, Lean would still accept the proofs.
Exhaustive case-split over all elements of a finite type (`decide`/`native_decide`), ensuring no case is missed.	The identification of the finite type with its mathematical counterpart (e.g. that the 8-element type `E8MaxSub` faithfully represents Dynkin's classification of E_8 maximal subalgebras).
Arithmetic in $\mathbb{Q}(\sqrt{5})$ (the ring `QSqrt5`) and $\mathbb{Z}$-matrix algebra (`Mat8`).	The McKay correspondence theorem itself, the Dynkin classification theorem, and the Griess–Ryba classification. These are cited as established mathematical results and encoded as data, not proved from first principles in Lean.

Mitigation.

The risk of data-entry error is mitigated by three independent measures. (1) *Internal consistency checks*: Lean verifies $\sum d_i^2 = 60$ (A_5 order), $\sum d_i^2 = 120$ (2I order),

$\sum$ marks = 30 (Coxeter number), $\sum$ branch dims = 248, and Cayley–Hamilton for the E_8 Cartan matrix. A single mis-transcribed entry would generically violate at least one such check. (2) *Dual encoding*: the A_5 character table is independently encoded in both `FlavorCharTable.lean` and `A5CP.lean`, with no import dependency between them; agreement serves as a cross-check. (3) *Independent GAP/Python verification*: every hand-encoded datum is also verified by GAP 4.15.1 and/or Python scripts (Appendix B), providing a third implementation on a different code base.

What "63+ theorems" means.

The project contains approximately 200 Lean `theorem` declarations in total, including supporting lemmas, `@[simp]` rewriting rules, and structural bookkeeping (e.g. `file_integrity : True := trivial`). The "60+ named theorems" cited in this paper refer to the M01–M63 series and their direct corollaries—those with substantive mathematical content corresponding to numbered results in the text. Three declarations (`M57_rho4_integer_char`, `M57_rho5_integer_char`, `file_integrity`) have the trivial statement `True` and serve only as registry markers; they are not counted among the substantive theorems.

A.3 Main files and correspondence theorems

Table A3 Main Lean files and their theorem coverage.

File	Main theorems	Lines	Technique
QSqrt5.lean	$\mathbb{Q}(\sqrt{5})$ construction	∼200	constructive
FlavorCharTable.lean	M01–M03, M11, M28	∼140	`native_decide`
FlavorCore.lean	CG tensor construction	∼100	constructive
FlavorGalois.lean	M07, M19, M29, M30	∼220	`native_decide`
A5CP.lean	M49, M50, M53–M57	∼350	`decide`/`native_decide`
E6Complex.lean	M51 (27 complex type)	∼200	`decide`
E8MaxSub.lean	M58	∼200	`decide`
E8BlowUp.lean	M59–M61	∼350	`decide`
E8Cartan.lean	Cartan matrix eigenvalues	∼200	`decide`
E6Trinification.lean	M62–M63, Z_3 structure	∼270	`decide`
G1SelCore.lean	Consequence layer	∼310	constructive

A.4 Overview of the verification method

The three main proof methods used in this project are:

`decide`: Fully automated proof for decidable predicates. The truth value is determined by exhaustive traversal on a finite type. The main theorems of E8MaxSub.lean (8 candidates), E8BlowUp.lean (9 nodes), E6Trinification.lean (6 candidates), and E6Complex.lean (M51) are all closed by this method.

`native_decide`: The native code version of `decide`, evaluating using compiled code rather than the Lean kernel evaluator. Used in character table calculations involving $\mathbb{Q}(\sqrt{5})$ arithmetic, such as FlavorCharTable.lean and A5CP.lean.

Constructive proof: General type-theoretic methods (`intro`, `apply`, `exact`, `simp`, `ring`, etc.) used in QSqrt5.lean (field structure of $\mathbb{Q}(\sqrt{5})$) and AddressPrinciple.lean (Paper I main theorem).

A.5 Dependencies

The dependencies between the main files are:

```
QSqrt5.lean
  +-- FlavorCharTable.lean
  |     +-- FlavorGalois.lean (M07, M19, M29, M30)
E6Complex.lean (M51, independent)
  +-- A5CP.lean (M49, M50, M53-M57)
E8MaxSub.lean (M58, independent)
E8BlowUp.lean (M59-M61, independent)
E8Cartan.lean (independent)
E6Trinification.lean (M62-M63, independent)
FlavorCore.lean (independent)
G1SelCore.lean (independent)
```

A5CP.lean defines the necessary character table data for M49–M57 in a self-contained manner and has no import relationship with FlavorCharTable.lean, constituting independent dual validation. M51, M58, M59–M61, M62–M63, and FlavorCore are all independent files, structurally guaranteeing the independence of each theorem.

A.6 Reproduction steps

1. Install Lean 4 and Lake (Lean's build system).
2. Clone the repository: `git clone https://github.com/nuu/A5_SM_Pathway`
3. Execute `lake build`.
4. Type checking of all theorems is complete; confirm no `sorry`/axioms.

The Mathlib version is pinned in `lakefile.lean`.

Appendix B GAP verification scripts

B.1 Environment

Table B4 GAP verification environment.

Item	Value
System	GAP 4.15.1 (Windows)
Main package	CTblLib (Character Table Library)
Usage	Computational verification of character tables and representations

Verification using GAP is a computational method independent of Lean formal proofs, providing a double guarantee of reliability by verifying the same theorem with different implementations.

B.2 Script list

Table B5 GAP/Python verification scripts.

Script	Verification	Main output	Result
M50_M54_verify.gap	M50 (ν_2), M54 (initial)	FS = $[1, 1, 1, 1, 1]$	PASS
M54_corrected_v2.gap	M54 (GaloisCyc$(\cdot, 2)$)	σ permutation	PASS
M52a_verify_v2.gap	M52a (Sym$^3(27)$ inv.)	$\dim \mathrm{Inv} = 1$	PASS
B4_node_removal.gap	M59 ($\hat{E}_8$ node removal)	9 Dynkin types	PASS
M58_verify.py	M58 (selection + 248 branching)	Survivor = $[E_6 \times A_2]$	PASS

B.3 Technical notes

In the verification of M54, the initial script used GaloisCyc$(x, -1)$ (complex conjugate), which could not detect the Galois automorphism σ. We confirmed that GaloisCyc$(x, 2)$ — acting as $\zeta_5 \mapsto \zeta_5^2$ and inducing $\sqrt{5} \mapsto -\sqrt{5}$ — is the correct implementation, by applying it to $\sqrt{5} = \mathrm{E}(5) + \mathrm{E}(5)^4 - \mathrm{E}(5)^2 - \mathrm{E}(5)^3$.

M58_verify.py is a Python script that includes automatic node removal determination for affine $\hat{E}_8$ (Dynkin type classification by arm length analysis) and dimensional consistency checks for 248 branching, providing a third verification method independent of the GAP script.

B.4 Complementarity with Lean verification

The three methods—Lean, GAP, and Python—are based on different implementation foundations and provide independent confirmations. In particular, M58 undergoes triple validation using Lean (`decide`), GAP (character table + FS indicator), and Python (node removal + dimension check).

Appendix C Complete list of P-layer hypotheses

In this research program, 25 numbered P-layer hypotheses have been identified. In this paper, the former E1 has been separated into E_{1a} (spatial dimension) and E_{1b} (number of generations).

Table C6 P-layer hypotheses promoted to M-layer.

ID	Hypothesis	M-theorem	Lean verified
P-2a	Non-solvability required	Born law (Paper I)	✓
P-2b	A_5 uniqueness	Klein [9] + M01–M29	✓
P-3c	$E_8 \to E_6 \times A_2$	M58 (§4.1)	✓
P-3d	$\mathrm{SU}(3)_{\mathrm{fam}} = \rho_3'$	M49 (§2.5)	✓
P-5b	$\mathrm{CP} = \arg(\lambda_3)$	M52 (§3.3)	✓
P-9b	SO(3) finite subgroups	Klein [9]	—

Table C7 Observational-fact hypotheses in Σ_{min}.

ID	Hypothesis	Σ_{min} name
P-1a	$n = 3$ + spinors	E_{1a}
—	$N_{\mathrm{gen}} = 3$	E_{1b}
P-3a	Non-abelian strong force	E_2
P-3b	Asymptotic freedom	E_3

Table C8 Structural P-layer hypotheses in Σ_{min}.

ID	Hypothesis	Σ_{min} name	Rationale
P-4a′	Equal-rank democracy	P-4a′	Physical adoption of T2
P-4b	Color identification	P-4b	Z_3 physical selection

C.1 M-promoted (not included in Σ_{min})

C.2 Observational facts (included in Σ_{min})

C.3 Structural P (included in Σ_{min})

C.4 Outside the scope of Paper IV

14 additional hypotheses (P-3e, P-5c, P-6, P-6b–d, P-7a–d, P-8a–c, P-9a) belong to other papers (Paper III, Paper V, G1′ programme) and do not affect Σ_{min}. See Table C9 for the complete list.

C.5 Aggregation

The completeness of Σ_{min} is guaranteed by the fact that, excluding the 6 M-promoted and 14 out-of-scope hypotheses from the 25 numbered ones, and adding E_{1b} (explicitly revealed by the separation of the old E1), 6 hypotheses (4 **[E]** + 2 **[P]**) cover all steps of the path (Table 15). Irreducibility is guaranteed by the fact that removing any one of the 6 hypotheses renders the path indeterminate (§6.3).

Table C9 P-layer hypotheses outside the scope of Paper IV.

ID	Hypothesis	Affiliation
P-3e	E_6 branch is physical	Paper IV §7
P-5c	$\delta_{\mathrm{CP}} \approx 70^\circ$	Paper V
P-6	τ-preceding condensation	Paper III
P-6b	r_c is small	Paper III
P-6c	$\xi \approx 0.13$	Paper III
P-6d	$\theta_{23} \approx 13^\circ$	Paper III
P-7a	A_5 = singularity information source	G1′
P-7b	Area $\approx 1/g^2$	G1′
P-7c	$\beta^* \leftrightarrow$ Kähler	G1′
P-7d	4.33 TeV	G1′
P-8a	Cubic Yukawa domination	Paper III
P-8b	Channel II = residual mass	Paper III
P-8c	Mass hierarchy correspondence	Paper V
P-9a	Identifying "natural occurrences"	Whole program

Table C10 Summary of P-layer hypothesis classification.

Category	Count	In Σ_{min}?
M-promoted	6	No (established as theorems)
Observational facts **[E]**	3	Yes (E_{1a}, E_2, E_3)
Structural P	2	Yes (P-4a′, P-4b)
Outside scope	14	No (other papers)
Total (numbered)	25	—
Added decomposition	+1	E_{1b} ($N_{\mathrm{gen}} = 3$)
Total Σ_{min}	**6**	4 **[E]** + 2 **[P]**

References

[1] McKay, J.: Graphs, singularities, and finite groups. In: The Santa Cruz Conference on Finite Groups. Proc. Sympos. Pure Math., vol. 37, pp. 183–186. Amer. Math. Soc., Providence, RI (1980)

[2] Gonzalez-Sprinberg, G., Verdier, J.-L.: Construction géométrique de la correspondance de McKay. Ann. Sci. École Norm. Sup. **16**, 409–449 (1983) https://doi.org/10.24033/asens.1454

[3] Serre, J.-P.: Complex Semi-simple Lie Algebras. Benjamin, New York (1966)

[4] Numagaki, M.: Uniqueness of icosahedral holonomy from three physical principles and its falsifiable prohibition structure for fundamental constants. Found. Phys. (2026). Submitted. Zenodo DOI: 10.5281/zenodo.18801089

[5] Kronheimer, P.B.: The construction of ALE spaces as hyper-Kähler quotients. J. Differential Geom. **29**, 665–683 (1989) https://doi.org/10.4310/jdg/1214443066

[6] Gross, D.J., Harvey, J.A., Martinec, E., Rohm, R.: Heterotic string theory. Nucl. Phys. B **256**, 253–284 (1985) https://doi.org/10.1016/0550-3213(85)90394-3

[7] Dynkin, E.B.: Semisimple subalgebras of semisimple Lie algebras. Mat. Sbornik **30(72)**, 349–462 (1952). English translation: Amer. Math. Soc. Transl. Ser. 2, **6**, 111–244 (1957)

[8] Glashow, S.L.: Trinification of all elementary particle forces. In: Kang, K., *et al.*(eds.) Fifth Workshop on Grand Unification, pp. 88–94. World Scientific, Singapore (1984)

[9] Klein, F.: Lectures on the Icosahedron and the Solution of Equations of the Fifth Degree. Teubner, Leipzig (1884)

[10] Feruglio, F.: Pieces of the flavour puzzle. Eur. Phys. J. C **75**, 373 (2015) https://doi.org/10.1140/epjc/s10052-015-3576-5

[11] King, S.F., Luhn, C.: Neutrino mass and mixing with discrete symmetry. Rep. Prog. Phys. **76**, 056201 (2013) https://doi.org/10.1088/0034-4885/76/5/056201

[12] Ma, E., Rajasekaran, G.: Softly broken A_4 symmetry for nearly degenerate neutrino masses. Phys. Rev. D **64**, 113012 (2001) https://doi.org/10.1103/PhysRevD.64.113012

[13] Altarelli, G., Feruglio, F.: Tri-bimaximal neutrino mixing, A_4 and the modular symmetry. Nucl. Phys. B **741**, 215–235 (2006) https://doi.org/10.1016/j.nuclphysb.2006.02.015

[14] Lam, C.S.: Determining horizontal symmetry from neutrino mixing. Phys. Rev. Lett. **101**, 121602 (2008) https://doi.org/10.1103/PhysRevLett.101.121602

[15] Luhn, C., Nasri, S., Ramond, P.: Simple finite non-Abelian flavor groups. J. Math. Phys. **48**, 123519 (2007) https://doi.org/10.1063/1.2823978

[16] Ramond, P.: Algebraic Dreams. hep-ph/0112261 (2001)

[17] Griess, R.L. Jr., Ryba, A.J.E.: Finite simple groups which projectively embed in an exceptional Lie group are classified! Bull. Amer. Math. Soc. **36**, 75–93 (1999) https://doi.org/10.1090/S0273-0979-99-00771-5

[18] Dechant, P.-P.: Clifford algebra unveils a surprising geometric significance of quaternionic root systems of Coxeter groups. Adv. Appl. Clifford Algebras **23**, 301–321 (2013) https://doi.org/10.1007/s00006-012-0371-3

[19] Moura, L., Ullrich, S.: The Lean 4 theorem prover and programming language. In: Automated Deduction – CADE 28. Lecture Notes in Comput. Sci., vol. 12699, pp. 625–635. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-79876-5_37

[20] The mathlib Community: The Lean mathematical library. In: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), pp. 367–381. ACM, New York (2020). https://doi.org/10.1145/3372885.3373824

[21] Hales, T., *et al.*: A formal proof of the Kepler conjecture. Forum of Mathematics, Pi **5**, 2 (2017) https://doi.org/10.1017/fmp.2017.1